UNIVERSITY AT ALBANY
State University of New York

College of Arts and Sciences
Department of Mathematics and Statistics

Applied Topology in Albany (ATiA) Seminar

Gonzalo Ortega Carpintero
City St Georges, University of London

Persistent fundamental groups and lower-central-quotient barcodes

Friday, November 13, 2026
12:00 p.m. Hudson 0110

Abstract. We present an efficient algorithm for computing the persistent fundamental group of a filtered CW-complex in terms of group presentations. From these presentations, we derive persistent invariants associated with the lower central series and obtain a family of barcodes that capture topological features not detected by persistent homology. This framework extends the scope of topological data analysis by enabling the study of non-abelian aspects of the fundamental group from filtered data.